

Bayram Bayramov

📍 Baku, Azerbaijan ✉ bayramovb578@gmail.com 🌐 github.com/baiiraam 🌐 bayram-bayramov 🌐 baiiraam

Profile

AI Engineer and Data Scientist with hands-on experience building machine learning and generative AI systems across the full lifecycle—from experimentation to deployment. Developed RAG applications and forecasting pipelines in a bootcamp setting, using PyTorch, Hugging Face, Docker, Airflow, Kafka, and vector databases. Evaluated model performance against real-world data during internships, diagnosing failure modes through systematic error analysis. Experience teaching and mentoring technical concepts, with strong communication skills for bridging AI/ML concepts to diverse audiences. Comfortable working in collaborative, fast-moving environments.

Tech Stack: PyTorch, TensorFlow, Keras, Hugging Face Transformers, RAG pipelines, LLM frameworks, Vector databases, Docker, Airflow, Kafka, Spark, Python, FastAPI, OpenCV, Git, PostgreSQL, Power BI, Tableau, NumPy, Pandas

Experience

DataMinds Azercell, Data Analytics and AI Bootcamp

Baku, Azerbaijan
Jan 2025 – Dec 2025
1 year

- Built and deployed a RAG application and forecasting pipeline during an intensive bootcamp, using Docker, Airflow, Kafka, and Spark alongside vector databases and LLM frameworks
- Implemented LLM-based retrieval using vector databases and open-source frameworks, building end-to-end systems from data ingestion to inference
- Gained practical experience with MLOps tools including pipeline orchestration, containerization, and stream processing

Kapital Bank, Data Science and AI Intern

Baku, Azerbaijan
Jan 2024 – Jan 2025
1 year 1 month

- Built Python tooling including pydub, audiossegment, and wave to segment, validate, and visualize audio data, supporting evaluation of an Automatic Speech Recognition (ASR) model
- Standardized data preparation workflows for a 4-person labeling team, improving consistency and turnaround for model testing
- Collaborated with engineers to align data processing with model requirements, contributing to more reliable ASR evaluation

Div Academy, Teacher and Tutor

Baku, Azerbaijan
Jan 2024 – Jan 2025
1 year 1 month

- Taught Python, SQL, PyTorch, TensorFlow, FastAPI, and Hugging Face Transformers, translating complex ML/AI and deep learning concepts into accessible practical knowledge
- Provided ongoing technical mentorship and code review, helping learners build skills in model design, training, and deployment
- Developed curriculum materials and coding exercises focused on real-world application

Azercosmos, Machine Learning Intern

Baku, Azerbaijan
Jan 2023 – Jan 2024
1 year 1 month

- Built and trained SAR image segmentation models including U-Net, Attention U-Net, and LRAU-Net in PyTorch, covering the full experimental lifecycle from data preprocessing to evaluation
- Diagnosed model failures through systematic analysis including failure-case visualization, per-region error statistics, and NDVI/NDBI indices
- Developed an evaluation framework that informed data relabeling and model improvements

Projects

AI Food Analyzer	Aug 2025 – Aug 2026
• Built a production-ready web application using FastAPI that identifies ingredients from meal photos and calculates nutritional information via Vision Language Models (OpenAI, Anthropic, Google Gemini) with automatic failover • Implemented 3-layer caching (VLM, nutrition, HTTP) achieving 500-600x speed improvement and parallel nutrition lookups with asyncio for up to 13x faster processing • Containerized with Docker and Docker Compose using multi-stage builds reducing image size to ~585MB, with GitHub Actions CI achieving 86% test coverage across 190+ tests • Integrated PostgreSQL with asyncpg for persistent storage and built both REST API and CLI interfaces with comprehensive OpenAPI documentation • Designed system architecture with file validation, SHA-256 hashing for caching, TTL-aware cache backends (JSON/SQLite), and structured logging with Loguru	

Education

Certificate Program	AI Academy , Artificial Intelligence • Studying machine learning and deep learning with a theory-heavy curriculum, implementing concepts through hands-on coding projects • Built an AI Food Analyzer project applying ML/DL techniques up to CNNs, with the full implementation detailed on GitHub	Baku, Azerbaijan Aug 2025 – Aug 2026
Master	Azerbaijan State University of Economics , Business and Data Analytics • Combining data science techniques with business strategy and decision-making • Studying advanced analytics, data visualization, and quantitative methods for business applications	Baku, Azerbaijan Sept 2025 – present
Bachelor	Baku Higher Oil School , Chemical Engineering • Built a strong foundation in mathematics, physics, and computational methods through a rigorous engineering curriculum • Developed analytical problem-solving skills applicable to data science and modeling	Baku, Azerbaijan Sept 2020 – June 2025